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CTI-ENGINE: An open-source cyber threat intelligence engine using advanced optimization and deep learning techniques for OSINT analysis

Abstract

6 In today's cyber landscape, the proliferation of Open-Source Intelligence (OSINT) sources such as social media, blogs, forums, and public threat reports offers a vast reservoir of data to identify emerging cyber threats. However, the unstructured, noisy, and high-dimensional nature of this data presents significant challenges in extracting relevant threat indicators in a timely and reliable manner. Traditional cyber threat intelligence (CTI) systems often lack efficiency and adaptability when dealing with such dynamic OSINT streams, especially in real-time threat detection and contextual understanding. To address these gaps, this study proposes CTI-ENGINE, an open-source cyber threat intelligence framework that employs a dual-stage intelligence pipeline for automated OSINT analysis. The primary objective is to enhance threat extraction accuracy, reduce computational complexity, and support real-time cyber threat awareness. The system integrates the enhanced Tasmanian devil optimization (ETDO) algorithm for optimal feature subset selection, eliminating redundant or irrelevant patterns from high-dimensional text data. These refined features are then processed using a coherent feed-forward quantum neural network, which ensures improved learning convergence and precise classification of threat-related entities such as attack sources, techniques, targets, and timestamps. The proposed system was empirically evaluated on a benchmark dataset comprising 214 real-world cyber incidents from 2016 to 2023, yielding over 1400 enriched threat instances. 14 Performance metrics demonstrated a precision of 97.974%, recall of 98.400%, and an F1-score of 98.186%, outperforming traditional LLM-based systems and rule-based OSINT tools. These results validate the CTI-ENGINE as a scalable, accurate, and analyst-friendly solution for modern CTI workflows driven by OSINT.

Keywords: cyber threat intelligence, open-source intelligence, threat detection, deep learning, feature analysis

1. Introduction

Cyberattacks are becoming increasingly frequent and complex in the digital age [1]. Cyber dangers are growing in size and sophistication, from ransomware to data breaches and state-sponsored intrusions. Organizations and governments are under constant pressure to secure their key infrastructure and sensitive data from a multiplicity of adversaries with diverse intentions and capabilities [2]. Cybersecurity teams may prepare for attacks faster by collecting and evaluating threat actor, attack vector, and vulnerability data. However, gathering high-quality cyber threat intelligence is difficult. Underground forums, social networks, blogs, and discussion boards provide huge, unstructured, and dynamic data [3]. The time and effort needed to manually parse and validate this information makes old approaches useless against contemporary threats. Intelligent and automated systems that can gain insights from huge and complicated datasets are needed to outperform attackers [4] which are capable of not just identifying important threat indicators but also giving context to enable educated decision-making. Open-source intelligence (OSINT) can be collected, processed, and analyzed to reveal national security, corporate risk, and cyber defense insights [5]. Internet protocol addresses, malicious software signatures, website URLs, exploit references, and attack technique talks are examples of cybersecurity data. Such information is stored on news websites, blogs, and social media. OSINTs accessibility and real-time insight are its main benefits [6]. Threat actors typically discuss methods or publish indicators of compromise in public forums before launching actual attacks.

Early threat detection allows security professionals to respond faster and more strategically by monitoring these channels. Unstructured, contextually dependent data may contain error or irrelevant noise. Signals from background babbling are hard to distinguish. Extracting and interpreting OSINT sources are difficult due to their linguistic and geographical distribution [7]. Thus, OSINT demands algorithms that understand plain English, validate data, and spot cyber threats. Cyber threat intelligence (CTI) uses Artificial intelligent (AI) techniques to assess OSINT could close this gap and increase threat exposure [8]. AI techniques can detect, categorize, and learn from complex patterns, transforming cybersecurity [9]. For open-source cyber threat intelligence extraction, this is crucial. Deep learning (DL) techniques, convolutional neural network (CNN) and attention models can identify threats in massive datasets [10]. These indications include Internet Protocol (IP) addresses, domain names, file hash values, viral references, and software flaw identifiers. Due to contextual links and language patterns, these approaches accurately extract information about threats from noisy text. Optimization techniques

can optimize hyper-parameters, reduce redundancy, and rank threat indicators by risk [11]. DL techniques and efficiency enable CTI engines to scale and adapt [12] which allow systems to learn from shifting hazard patterns, minimize false positives, and predict properly without human intervention. CTI awareness is rising, yet many technologies and platforms fall short [13][14].

Businesses often use proprietary systems with limited openness, flexibility, and scalability. These solutions may be difficult to integrate into infrastructure or adapt to threat environments. Many platforms tend to adapt to new methods of attack because they rely on recognized threats or data sources [15]. They struggle to process OSINT-rich natural language which makes harder to recognize new threats or predict attackers. Existing technologies are expensive and resource-intensive for smaller companies, academic institutions, and NGOs. A limited framework and lack of centered on communities development hinder collaboration and fast feature or analytical skill generation [16]. These restrictions emphasize the necessity for an open, flexible, and intelligent cyber threat intelligence solution. A freely accessible and expandable system with sophisticated learning methods, optimization strategies, and real-time data ingestion can help more stakeholders improve their cybersecurity [17]. An open-source CTI platform has various advantages over proprietary ones [18]. Developers and security professionals worldwide can improve the system, enhance its capabilities, and align it with real-world concerns. Auditing and customizing the system improves data processing and connection with internal tools for organizations [19]. Building a strong and cooperative cybersecurity environment requires interoperability and standardization, which open-source development promotes. Data extraction, classification, and prioritizing can be automated using machine learning models, relieving analysts. Optimization algorithms can give threat intelligence quickly and efficiently as data volumes expand [20]. Despite the increasing reliance on OSINT for CTI, existing systems continue to face significant limitations in processing and leveraging the vast, dynamic, and unstructured data derived from sources such as social media, blogs, and underground forums. One of the primary challenges is the high-dimensional and noisy nature of OSINT data, which often contains redundant, irrelevant, or misleading information that traditional CTI frameworks struggle to filter and analyze effectively. Many current systems lack real-time processing capabilities, making them reactive rather than proactive in detecting emerging threats. This latency in response time severely limits their usefulness in fast-evolving cyber threat landscapes. Furthermore, rule-based and static large language model (LLM)-driven approaches tend to rely

heavily on pre-defined indicators, which hampers their ability to generalize to new, evolving, or zero-day attack patterns. The lack of adaptability to novel threats makes these systems less effective in modern threat environments. Another notable shortcoming is the limited scalability and openness of existing proprietary CTI solutions, which are often expensive, rigid, and unsuitable for integration into diverse cybersecurity ecosystems—posing a particular challenge for small organizations, academic institutions, and non-profits. Compounding these issues is the inefficient handling of features in large-scale text-based data, where the absence of effective optimization strategies leads to increased computational costs and reduced accuracy. In addressing these limitations, several recent works have explored **the application of deep learning and natural language processing (NLP) techniques** for CTI. However, they too exhibit critical gaps. Many of these models fail to incorporate effective feature selection optimization, resulting in longer training times and a higher risk of overfitting. Moreover, a significant number of these models lack the ability to semantically interpret contextual threat indicators, which negatively impacts the precision and recall of extracted information. Integration with real-time, streaming OSINT data remains underdeveloped, limiting the timeliness and relevance of the intelligence produced. The absence of open-source frameworks that encourage community-driven contributions continues to impede collaborative development, standardization, and the rapid enhancement of CTI capabilities across different stakeholders. To address the above issues, this study introduces CTI-ENGINE, an open-source and scalable cyber threat intelligence platform that leverages AI-driven techniques in a dual-stage processing pipeline:

1. Enhanced Tasmanian devil optimization (ETDO) algorithm handles the high-dimensional nature of OSINT data by performing optimal feature subset selection. ETDO eliminates irrelevant and noisy features, drastically reducing computational complexity and improving the quality of extracted threat indicators.
2. Coherent feed-forward quantum neural network (CFF-QNN) is used to enhance learning efficiency and classification accuracy. The model processes refined features to identify key threat elements such as attack sources, methods, targets, and timestamps, ensuring better convergence and lower misclassification rates.
3. The framework is tested on a real-world benchmark dataset encompassing 214 cyber incidents (2016–2023), extracting over 1400 enriched threat instances.

8 The rest of this paper is organized as follows. Section 2 presents the review of literature on open-source cyber threat intelligence frameworks using DL models. 17 The detailed working process of proposed work is present in Section 3 with the feature analysis and detection phase. 9 The results and comparative analysis present in Section 4. Finally, the paper concludes in Section 5.

2. Related works

18 Balasubramanian et al. 2025 [21] launched TSTEM, a cloud-based, real-time platform that extracts cyber threat intelligence (CTI) from freely accessible sources to improve IT and OT cybersecurity. CTI is essential for early detection of threats, yet most businesses struggle to acquire relevant intelligence. TSTEM's containerized micro service architecture uses Tweepy, 6 Scrapy, Terraform, Elastic search, Logstash, Kibana (ELK), Kafka, and MLOps. Custom web crawlers, a two-level classification pipeline, and advanced NLP models for categorization and entity extraction efficiently identify indicators of compromise (IOCs). This study provides an end-to-end, scalable, and independent solution for real-time CTI detection, searching, and 11 sharing from different online sources, unlike previous research that concentrated on isolated obtaining components. Addison et al. 2025 [22] addressed smart home cybersecurity issues caused by IoT device adoption. Traditional security strategies using static methods are useless against emerging threats. A proactive, low-cost Raspberry Pi solution which combines real-time threat information from the malware information sharing platform (MISP) with Suricata-based intrusion prevention and detection systems is investigated. The system detected and prevented 99.9% of threats, exceeding conventional methods by adapting faster. IoT security has advanced with this practical, adaptable method that uses real-time threat intelligence to secure smart homes. Arikkat et al. 2025 [23] introduced SeCTIS, a secure cyber threat intelligence sharing (CTI) platform to address the escalating cybersecurity risks faced by interconnected, IT-dependent enterprises. With Swarm Learning and Blockchain, SeCTIS allows information about threats sharing while protecting data. Numerous trials show that SeCTIS is accurate and reliable, withstanding data or model security attacks. This unique approach improves threat intelligence sharing and creates a secure and trustworthy cooperation environment, advancing organizational cybersecurity strategy. Jamil et al. 2025 [24] launched iMTOSA, an intelligent resource allocation system for IoE-based edge networks, especially 6G technologies. As IoE devices connect and share computing power at the edge, efficient planning of resources is essential for

excellent QoS and network usage. Based on extensive simulations, iMTOSA obtained task success rates of 89%–99.8%, compared to 30%–40% for non-intelligent. Using PERT and the central limit theorem to assess task fulfillment reliability, the framework worked well in industrial and corporate settings. As a whole, iMTOSA improves task effectiveness and efficacy in 6G IoE edge networks. Luh et al. 2025 [25] created PenQuest, a gamified meta-model that incorporates MITRE ATT&CK, D3FEND, CAPEC, and NIST SP 800-53 cybersecurity. PenQuest improves threat evaluation and practitioner training using game theory and cybersecurity. Avrahami et al. 2025 [26] investigated how CTI integrates technology and human components to improve cyber resilience. The study analyzes associations using linear regression and PROCESS on 208 cybersecurity-aware adults, including 131 professionals. The study emphasizes merging CTI with human-centric techniques to generate proactive, adaptive defense mechanisms. To increase long-term security and confidence from stakeholders, it advises adding threat intelligence to ISO 27001 cybersecurity requirements.

Dimitriadis et al. 2025 [27] presented EVACTI for assessing CTI products actionability across the CTI sharing lifecycle—production, propagation, and consumption. CTI improves cybersecurity however current evaluation methods generally use isolated criteria and are subjective. EVACTI uses ENISA-standardized actionability criteria to fill these gaps with a uniform and complete assessment. It considers CTI producers' and customers' operational contexts for greater precision evaluations and practical uses. EVACTI improves communication and intelligence priority by incorporating effectiveness into the CTI lifecycle through a specialized CTI object. This method motivates manufacturers to enhance CTIP quality prior sharing and lets consumers make educated choices, thus improving cybersecurity and threat intelligence sharing. Chatziamanetoglou et al. 2025 [28] developed a complete approach for assessing CTI quality and operational relevance to improve cybersecurity risk management. The quantitative techniques to evaluate CTI are needed as its volume, speed, and complexity rise. Using the MISP data model, the study provides explicit quality measures using formulas from mathematics and provides subjectively fuzzy-Analytic Hierarchy Process and objective entropy-based weighting algorithms. Current CTI evaluation practices lack common metrics and connections with operating platforms like threat intelligence platforms. The proposed framework overcomes these issues. A proof of concept using genuine Open Source Intelligence data shows the framework's applicability and usefulness, improving CTI prioritizing and enterprise

protection and threat response methods. Cohen et al. 2025 [29] proposed a human-AI partnership to model cyber threat intelligence (CTI) and detect unusual communications that may indicate cyber events. In a Reciprocal Human-Machine Learning (RHML) approach, cybersecurity specialists and AI systems learn from each other through iterative interactions to improve threat identification. Concept maps show that detection performance and experts' danger landscape conceptualization increased with time. Human-AI collaboration can improve cybersecurity by providing a more dynamic, updated and reactive CTI strategy, improving operational performance in recognizing and reacting to advanced cyber attacks. Arikkat et al. 2025 [30] emphasized the possibilities of social media for meaningful cyber threat intelligence, notably IoC extraction. From enormous social media data, a convolutional neural network (CNN) identified important IoCs such IP addresses, URLs, file hashes, domain names, and CVE IDs with an F1-score of 98.80% and a detection rate of 99.65%. Most often, URLs were shared, with 48.67% genuine threats. XGBoost was the best bot detection model, with macro and weighted F1-scores of 0.814 and 0.925, respectively. These results show that social media platforms are more important in providing fast, accurate threat information and helping cybersecurity experts detect and react to developing cyber threats.

Problem Definition: Despite significant progress in the domain of CTI, numerous challenges (Table 1) persist in transforming unstructured OSINT data into actionable intelligence. For instance, TSTEM [21] and SeCTIS [23] introduced scalable architectures and secure sharing mechanisms, focusing on real-time intelligence extraction and collaborative threat sharing, respectively. However, these systems often rely on complex deployment infrastructures or assume trust among stakeholders, making them less adaptable for lightweight, community-driven CTI solutions. Similarly, [22] proposed real-time threat prevention in IoT contexts using threat feeds and Suricata-based detection, but such setups are tightly coupled with specific environments and lack generalization across broader OSINT sources. While [24] and [25] address edge computing and training aspects in CTI, they do not directly tackle the core problem of processing noisy, high-volume OSINT data for threat detection. A major research focus has also been on evaluating CTI quality and actionability, as seen in [27] and [28], which emphasize the need for standardization and measurable threat intelligence metrics. However, these works largely assume that the CTI data is already well-structured and filtered, overlooking the challenge of extracting meaningful information from raw OSINT streams.

Table 1 Summary of recent research in OSINT-based CTI—methodologies, key findings, and identified research gaps

Ref	Methodology	Technique Used	Dataset	Findings	Research Gaps
[21]	Platform development	TSTEM with NLP, micro-services, ELK, Kafka, MLOps	Open CTI sources	98%+ accuracy in IOC detection with low false positives	Limited external platform integration
[22]	Experimental simulation	MISP + Suricata on Raspberry Pi	Simulated attacks	99.9% detection/prevention rate in smart homes	Scalability and broader applicability not evaluated
[23]	Framework design	SeCTIS with Blockchain, Swarm Learning, Zero Knowledge Proofs	CTI sharing scenarios	Secure, privacy-preserving CTI sharing	Practical scalability and deployment validation needed
[24]	Simulation-based	iMTOSA with intelligent scheduling & central resource control	IoE-based edge networks	89%–99.8% task success rate	Needs testing in real-time dynamic environments
[25]	Gamified model design	PenQuest with MITRE, D3FEND, NIST, Game Theory	Simulated IT/OT cyber battles	Effective for practitioner training and risk evaluation	Improvement needed in threat likelihood/detection modeling
[26]	Quantitative survey + analysis	CTI impact on human and org. resilience	208 respondents (131 pros)	Positive link between CTI, human factors, and resilience	Behavioral adaptation not a significant mediator
[27]	Lifecycle-based evaluation	EVACTI with ENISA actionability standards	CTI lifecycle stages	Standardized, contextual CTIP evaluation	Broader testing across CTI platforms required
[28]	Metric-based evaluation	MISP model + AHP + Entropy weighting	Open-source threat data	Effective CTI quality scoring for prioritization	Lack of integration with real-time TIPs
[29]	Human-AI interactive learning	RHML with expert-AI iterations	6,651 hacker forum posts	Improved expert awareness & detection accuracy	Scope limited to textual, forum-based threat intelligence
[30]	ML-driven IoC extraction	CNN for IoCs, XGBoost for bot detection	Social media data	98.8% F1 score, URLs as top IoC	Lacks real-time deployment evaluation and integration

The recent efforts like [30] demonstrated the value of social media platforms for extracting IoCs using CNNs, achieving high detection rates. Yet, the reliance on single-task models limits their adaptability to diverse CTI requirements such as contextual understanding, source attribution, and temporal correlation of threats. The increasing interest in using LLMs for CTI is evident in [29] and [31], where LLM-based chat-bots such as GPT-4, GPT4all, Vicuna, and Falcon were tested on classification tasks for cyber threat awareness. Although **GPT-4 achieved** a commendable **F1-score of 0.94** in classification, all evaluated LLMs fell short in effectively extracting threat entities compared to specialized models. The lack of domain-specific optimization, inability to handle noisy contextual data, and limited interpretability reduce their practical value in real-time CTI workflows. These studies highlight persistent gaps in current CTI systems: inadequate semantic understanding of noisy OSINT data, lack of effective feature optimization methods, underutilization of real-time processing pipelines, limited open-source adaptability, and the performance limitations of general-purpose LLMs for cybersecurity entity recognition. Thus, there is a pressing need for a robust, open-source CTI framework that can efficiently extract, filter, and classify cyber threat indicators from unstructured OSINT in real time, while overcoming the computational and contextual limitations of existing models.

3. Materials and Methods

Fig. 2 illustrates the architecture of the proposed CTI-ENGINE framework, designed to extract actionable CTI from OSINT data. The process begins with the collection of real-world cyber incident data from publicly available OSINT platforms such as Twitter, blogs, forums, pastebins, and news feeds. These raw text sources are acquired through crawling or streaming mechanisms and then passed through a text normalization phase where irrelevant characters, noise, and formatting issues are cleaned to prepare the corpus for downstream processing. The normalized text is then converted into a structured representation through text Vectorization using BERT, a pre-trained contextual embedding model. To enhance contextual relationships and capture patterns in the text, N-gram generation is applied. This results in high-dimensional feature vectors that represent semantic and syntactic characteristics of the source text. The next stage involves feature optimization using the enhanced Tasmanian devil optimization (ETDO) algorithm. This step reduces dimensionality and filters out irrelevant or redundant features, thereby improving the quality and performance of the classification model.

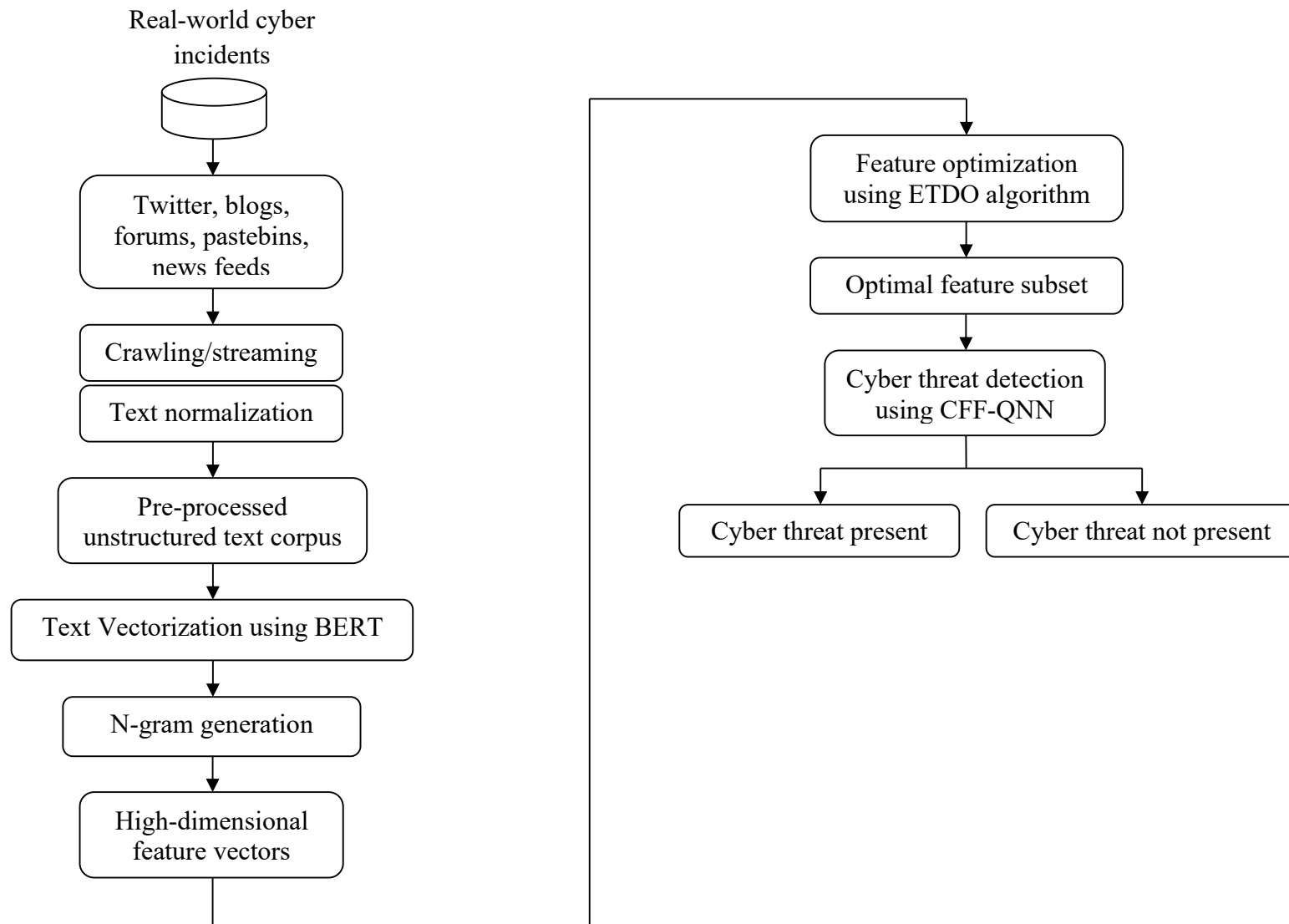


Fig. 1 The proposed CTI-ENGINE framework for cyber threat intelligence using OSINT data

The optimized feature subset is then fed into a custom feed-forward quantum neural network (CFF-QNN), which is trained to detect cyber threats by learning complex nonlinear patterns and relationships within the data. It produces the final classification results, distinguishing whether a cyber threat is present or not present. This binary output allows security analysts to focus on relevant and verified threat signals extracted from massive volumes of noisy and unstructured OSINT data.

3.1 Data collection

The proposed CTI-ENGINE framework was empirically evaluated using a custom-compiled benchmark dataset referred to as the Real-World OSINT CTI Corpus. This dataset comprises 214 verified cyber incidents that occurred between January 2016 and December 2023 [31]. Data was sourced from a broad range of open-source intelligence (OSINT) platforms, including Twitter, blogs, dark web forums, pastebin dumps, and cybersecurity news feeds. Data collection was carried out using a hybrid strategy combining streaming APIs (especially for Twitter) and custom-designed web crawlers for continuously scraping unstructured text from public feeds and repositories. The initial collection phase yielded over 4,700 raw text records, which were subsequently pre-processed and curated to produce 1,428 enriched threat intelligence instances. These enriched samples included detailed information such as attack sources, attack techniques, targeted entities, timestamps, campaign identifiers, and assigned severity levels. Each sample was annotated with indicators of compromise (IoCs), including IP addresses, domain names, malware hashes, and file signatures. In total, over 5,300 unique IoCs were extracted and verified using external intelligence sources such as ThreatMiner, AbuseIPDB, and VirusTotal. To support supervised learning, the dataset was labeled for both classification and named entity recognition (NER) tasks. Annotation was performed using a semi-supervised strategy, followed by manual verification by domain experts to ensure data quality and ground-truth reliability. Samples were stratified into training, testing, and validation splits, with 999 samples (70%) used for model training, 286 samples (20%) for testing, and 143 samples (10%) for validation. To mitigate class imbalance, especially within minority threat categories such as APTs or ransomware, Synthetic Minority Over-sampling Technique (SMOTE) was employed during the training phase. The dataset encompasses multilingual text, with English being the predominant language, while a minority of entries appeared in Russian, Chinese, and Arabic. Each threat instance was

vectorized using a combination of BERT-based contextual embeddings and N-gram statistical feature generation, resulting in an initial feature vector of approximately 1,024 dimensions per sample. A TF-IDF-based filtering mechanism was employed to eliminate noisy or redundant features, leading to the removal of approximately 6.2% of low-relevance records. Threat types were categorized into phishing, ransomware, DDoS, data leakage, malware campaigns, and advanced persistent threats (APTs). Severity levels were annotated on a four-point scale—Low, Medium, High, and Critical—based on contextual metadata and expert evaluation. The dataset was stored in structured CSV and JSON formats and maintained as a static batch resource, with periodic annual updates to include the most recent threat intelligence. The richness and heterogeneity of this dataset provided a robust foundation for training and evaluating the CTI-ENGINE framework across classification, threat extraction, and real-time decision-making tasks.

Table 2 Description of dataset with real-world cyber incidents from 2016 to 2023

Attribute	Description
Dataset name	Real-World OSINT CTI Corpus
Data sources	Twitter, blogs, forums, pastebins, cybersecurity news feeds
Collection method	Streaming APIs + custom web crawlers
Timeframe covered	January 2016 – December 2023
Number of cyber incidents	214 verified incidents
Total samples (threat instances)	1,428 enriched instances
Raw records collected	4,700+ unstructured text snippets
Samples used for training	999 samples (70%)
Samples used for testing	286 samples (20%)
Samples used for validation	143 samples (10%)
Languages present	English (major), minor entries in Russian, Chinese, Arabic
Threat categories	Phishing, ransomware, DDoS, data leak, malware, APTs
Severity labels	Low, Medium, High, Critical (based on expert annotation)
Indicators of compromise (IOCs)	IP addresses, hashes, domain names, filenames, malware signature
Structured threat entities	Attack source, target, technique, timestamp, severity, campaign ID
Number of IOCs extracted	5,300+ unique IOCs
Text format	Raw HTML, plaintext, JSON
Vectorization method	BERT embeddings + N-gram feature generation
Initial feature vector size	~1,024 dimensions per sample
Noise removal threshold	TF-IDF-based filtering, 6.2% noisy records removed

Annotation method	Semi-supervised with security expert verification
Class balance	Balanced with SMOTE during training
Data storage format	Structured CSV and JSON files
Dataset update frequency	Static batch (non-streaming, curated annually)
Ground truth verification	Cross-referenced with ThreatMiner, AbuseIPDB, and VirusTotal

3.2 Feature analysis

In the proposed CTI-ENGINE framework, feature analysis plays a critical role in transforming raw, unstructured text into a refined and highly informative representation suitable for deep learning-based threat intelligence extraction. This phase involves a multi-step pipeline comprising advanced Vectorization, statistical enrichment, and nature-inspired feature selection.

3.2.1 Text Vectorization using BERT

Initially, each textual instance—extracted from OSINT sources like tweets, forums, or blogs—is encoded using the bidirectional encoder representations from transformers (BERT) [32]. BERT provides contextualized word embeddings by understanding bidirectional relationships in text. This allows the model to generate rich, semantic representations of threat indicators, even when they appear in diverse linguistic contexts. For each sample, BERT generates a dense embedding vector (typically 768 to 1024 dimensions) that captures syntactic and semantic nuances essential for detect threat-specific terms like malware names, attacker aliases, and target organizations.

3.2.2 N-Gram Generation

To complement BERT embeddings with frequency-based lexical patterns, the system also incorporates N-gram feature generation [33]. These N-grams are extracted from the preprocessed tokens and help in capturing repetitive threat expressions, jargon, and pattern-based indicators that may not be effectively represented in contextual embeddings. For example, common N-grams like “ransom demand”, “data breach”, or “C2 server” often appear in cyber threat reports and serve as strong signals for classification. The N-gram features are combined with BERT embeddings, resulting in a high-dimensional and feature-rich representation of each instance.

3.2.3 Feature optimization

The enhanced Tasmanian devil optimization (ETDO) algorithm plays a crucial role in refining the high-dimensional feature space generated by BERT embeddings and N-gram representations. Typically, this combined feature vector—often exceeding 1,024 dimensions—contains redundant or irrelevant patterns that may hinder classification performance and inflate computational costs. ETDO, an improved version of the original Tasmanian devil optimization (TDO) algorithm [34], is introduced to address this challenge by performing optimal feature subset selection. While the standard TDO is inspired by the scavenging and adaptive hunting behaviours of Tasmanian devils, leveraging probabilistic exploration and exploitation strategies, it often suffers from premature convergence in high-dimensional and noisy data environments such as CTI text. ETDO enhances this foundation by incorporating adaptive local intensification using Lévy flight-based mutations to escape local optima, dynamic penalty weighting to encourage sparsity and reduce redundancy, and gradient-inspired directional updates that enable faster and smoother convergence. ETDO employs a binary encoding mechanism specifically designed for feature selection, where each bit represents whether a feature is selected or discarded. The inputs to the ETDO algorithm include the full BERT+N-gram feature vector per sample, a fitness function based on classification accuracy and parameters such as population size and maximum iterations. The Tasmanian devil's foraging is used to find a suitable food source, describes as follows.

$$AM = \begin{bmatrix} AM_1 \\ \vdots \\ AM_g \\ \vdots \\ AM_P \end{bmatrix}_{P \times Q} = \begin{bmatrix} am_{1,1} & \cdots & am_{1,K} & \cdots & am_{1,q} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ am_{1,1} & \cdots & am_{g,K} & \cdots & am_{g,q} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ am_{1,1} & \cdots & am_{p,K} & \cdots & am_{p,q} \end{bmatrix}_{P \times Q} \quad (1)$$

An initial Tasmanian devil inhabitants was referred to as AM, and the g-th probable solution was represented as AM_g . Also, the applicant's evaluation of k-th value was strong-minded as $am_{g,K}$, and the space comprise the factor and difficulty q is represent as P.

$$fit = \begin{bmatrix} fit_1 \\ \vdots \\ fit_g \\ \vdots \\ fit_P \end{bmatrix}_{P \times Q} = \begin{bmatrix} fit(Am_1) \\ \vdots \\ fit(Am_g) \\ \vdots \\ fit(Am_P) \end{bmatrix}_{P \times Q} \quad (2)$$

while f denotes the best solutions and fit_g determines the g -th candidate solutions of the main function. The Tasmanian devil from the g -th location selects a key carcass location in the neighborhood, which has a score that depends on the choice of written random conditions.

$$CN_g = Am_h, h \in \{1, 2, \dots, P \mid h \neq g\}, g = 1, 2, \dots, P \quad (3)$$

The Tasmanian devil's favorite scavenger hunt was identified CN_g based on carcasses selected from its foraging grounds. The new agent can reveal the updated location, as provided here.

$$am_{g,K}^{New,r_1} = \begin{cases} am_{g,K} + T.(C_{g,K} - H.a_{g,k}), & Jc_g < J_g \\ am_{g,K} + T.(C_{g,K} - a_{g,k}), & otherwise \end{cases} \quad (4)$$

$$Am_g = \begin{cases} Am_{g,K}^{New,r_1}, & J_g^{New,r_1} < J_g \\ Am_g, & otherwise \end{cases} \quad (5)$$

According to the master strategy model, a new and improved position is indicated $am_{g,K}^{New,r_1}$. Exploitation phase is similar to the number of other Tasmanian devils in selecting prey, based on the position of the predator.

$$X_g = Am_h, h \in \{1, 2, \dots, P \mid h \neq g\}, g = 1, 2, \dots, P \quad (6)$$

Once the enhanced effect is determined based on the base function, it modifies its current instance and assigns it to the new role as written.

$$am_{g,K}^{New,r_2} = \begin{cases} am_{g,K} + T.(C_{g,K} - H.a_{g,k}), & Jy_g < J_g \\ am_{g,K} + T.(a_{g,K} - m_{g,k}), & otherwise \end{cases} \quad (7)$$

$$A_g = \begin{cases} Am_{g,K}^{New, r_2}, & J_g^{New, r_2} < J_g \\ A_g, & otherwise \end{cases} \quad (8)$$

If the devil's value was chosen as the hunters' target and the previous value was chosen next to it, a better planning solution was implemented and the reconstruction of the devil's location in subsequent years was evaluated.

$$Am_{g,K}^{New} = am_{g,K} + (2t - 1)t.am_{g,K} \quad (9)$$

$$t = 0.01 \left(1 - \frac{1}{Maximumiteration} \right) \quad (10)$$

$$Am_g = \begin{cases} Am_g^{New}, & J_g^{New} < J_g \\ Am_g, & otherwise \end{cases} \quad (11)$$

The current incarnation is depicted as a single iteration, but the total number of iterations is represented as the maximum iteration. As a result of its new state, it is possible to select the most relevant and low-dimensional features in the database elements and determine the fitness value J_g , g -th, and Tasmanian devil neighborhood A_g^{New} . Algorithm 1 defines the working process of feature optimization by ETDO. Throughout the optimization process, candidate solutions—each shows unique feature subset—are evaluated, and low-performing features are systematically eliminated. The output is a reduced, optimized feature vector typically condensed from over 1,000 dimensions to around 300–400, significantly improving model generalization, reducing noise, and minimizing computational overhead. By enabling faster training and inference, enhancing classification accuracy, and mitigating overfitting, the ETDO algorithm substantially improves the robustness of the proposed CTI framework. Related to other optimization methods such as particle swarm optimization (PSO), genetic algorithm (GA), and even standard TDO, ETDO demonstrates superior performance in convergence speed and final classification quality, making effective for real-time OSINT-based cyber threat modelling.

Algorithm 1 Feature optimization using ETDO

Input : Initial population of feature masks, Maximum number of iterations, feature

vectors, fitness function, step size, Lévy flight control factor, penalty weight

Output : Optimal feature selection

1. Initialize the population with random binary vectors representing feature subsets. Assign each Tasmanian devil a candidate solution (feature mask) and define initial step sizes.
2. Evaluate the fitness of each devil using the classification accuracy of the quantum neural network on the selected feature subset.
 $CN_g = Am_h, h \in \{1, 2, \dots, P \mid h \neq g\}, g = 1, 2, \dots, P$
3. Update positions of devils using a balance of exploration and exploitation based on adaptive step sizes and Lévy flight mutation to escape local optima.

$$am_{g,K}^{New,r_1} = \begin{cases} am_{g,K} + T.(C_{g,K} - H.a_{g,k}), & Jc_g < J_g \\ am_{g,K} + T.(C_{g,K} - a_{g,k}), & otherwise \end{cases}$$
4. Apply dynamic penalty to solutions with redundant features, promoting sparse and relevant feature selection.
 $X_g = Am_h, h \in \{1, 2, \dots, P \mid h \neq g\}, g = 1, 2, \dots, P$
5. Modify candidate solutions based on observed performance, replacing weaker agents with stronger ones or mutated versions.

$$am_{g,K}^{New,r_2} = \begin{cases} am_{g,K} + T.(C_{g,K} - H.a_{g,k}), & Jy_g < J_g \\ am_{g,K} + T.(a_{g,K} - m_{g,k}), & otherwise \end{cases}$$
6. Reconstruct positions by updating each devil's location using the gradient-inspired search direction based on the current best-known solution.
 $Am_{g,K}^{New} = am_{g,K} + (2t - 1)t.am_{g,K}$
7. Identify and record the best-performing feature subset and associated fitness score.
8. Repeat steps 2–7 until the maximum number of iterations is reached or convergence criteria are met.
9. Return the best solution

3.3 Open-source cyber threat intelligence framework

The refined feature set obtained through the ETDO process is further processed using a coherent feed-forward quantum neural network (CFF-QNN), an architecture that integrates quantum-inspired computation with classical feed-forward learning techniques. This hybrid model is used to handle high-dimensional, sparse cyber threat intelligence data by transforming the optimized features into quantum-compatible representation. Unlike conventional neural network (CNN) [35] that operate purely on classical principles, the CFF-QNN leverages quantum coherence to maintain the superposition of input states across layers and faster convergence during training. The key inputs to the CFF-QNN include the reduced, binary-encoded feature vectors—typically ranging from 300 to 400 features—alongside labels that describe various threat entities such as

the attack source, technique used, targeted component, and timestamp of the incident. The internal design of the CFF-QNN employs quantum phase rotation in neuron encoding and a phase-adaptive back-propagation mechanism to optimize learning. CFF-QNN incorporates an entanglement-based gate mechanism that captures complex inter-feature relationships—critical for accurately classifying cyber threat contexts. The outputs of the model include multi-label predictions for structured threat intelligence entities such as the attack source, attack technique, target and timestamp of the threat. Each label is also associated with a classification probability score, improving interpretability. The layers of CFF-QNN contain of net of single-qubit and constrained revolution entries, whose parameters are separated according to their rotation angle. For reasons that will become clear soon, choose to achieve altogether revolutions along the y-axis. They are exactly labeled by a single-qubit spin gate, communicated by single-qubit calculations $\{|0\rangle, |1\rangle\}$ as follows.

$$r_q(\theta) = \text{Exp}\left(-h\theta\sigma_q/2\right) = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix} \quad (12)$$

Using the Pauli matrix σ_q . The rotation angle of the r_q gate operating on a preparatory state—which is assumed to be a generic state $r_q(n)|0\rangle$ —is recorded as the weighted data point in the first layer. Such an encoding may be carried out with a single single-qubit gate parameterised by $\theta = w = \sum_{h=1}^B P_h Z_h + n$ since all of the rotations are about the same axis:

$$r_q(P_B Z_B) \cdots r_q(P_2 Z_2) r_q(P_1 Z_1) [r_q(n)|0\rangle] = r_q(w)|0\rangle \quad (13)$$

To allow for nonlinear data processing in quantum circuits, this data encryption process can be repeated on many different qubits. A description of CFF-QNN with n qubits in the first layer is as follows.

$$|\psi\rangle = [r(w)|0\rangle]^{\otimes b} \quad (14)$$

To create new qubits using a parameterized rotational gate as dependent term $r_{q;b+1}(\theta)$ operating on the g -th qubit, and then use parameterized constrained rotational gates controlling the qubits in the first layer. For instance, the state of the first qubit controls the rotation that is given to the $(b + 1)$ -th qubit in order to connect the first node of the hidden layer to the first node of the second layer:

$$Dr_q^{1 \rightarrow b+1}(\theta) = |0\rangle_1 \langle 0| \otimes H_{b+1} + |1\rangle_1 \langle 1| \otimes r_{q;b+1}(\theta) \quad (15)$$

The additional nodes related by a similarly measured cyclic gate $Dr_q^{1 \rightarrow b+1}(\theta_{hg})$, each parameterized with independent weights (θ_{hg}) . CFF-QNN maintains the branches of the wave function corresponding to the control qubit in every state, measuring $|0\rangle$ and $|1\rangle$ without adjusting the state. For instance, the branch of the state when all of the control qubits are in state $|1\rangle$ applies the gate $r_{q;g} \left(\sum_{h=0}^b P_h \theta_{hg} \right)$ to qubit g , but the branch where all of the run qubits are in state $|0\rangle$ only applies the bias rotation to the target qubits.

$$u = \prod_{hg} Dr_q^{1 \rightarrow g}(\theta_{hg}) = \sum_P |P\rangle \langle P| \otimes_g r_{q;g} \left(\sum_{h=0}^b P_h \theta_{hg} \right) \quad (16)$$

The tensor product over g indicates that the rotation on the g -th qubit related with the branch of the wave function where the qubits are in state $|P\rangle$ is by an angle $\sum_{h=0}^b P_h \theta_{hg}$, matching the standard factor. In this case, $P = P_1, \dots, P_b$ is a moment filament with basics $P_h \in \{0, 1\}$, and the entire rounds ended all such filaments. We have included $P_0 = 1$ to represent the bias term. The system is prepared to have the procedure repeated in a later layer as all of the perception's output values are essentially maintained coherently.

$$\tilde{q} = \begin{cases} 1, & x(1) > x(0) \Leftrightarrow w > \alpha \\ 0, & p(1) < x(0) \Leftrightarrow w < \alpha \end{cases} \quad (17)$$

As in a traditional NN, we establish the class of the data point by performing a measurement and calculating the predicted value of an operator after all intermediate (hidden) layers. To produce a binary effect, the measurement result is passed into a classical nonlinear activation function σ .

$$\tilde{q} = \begin{cases} 1, & \sigma(\langle W_B \rangle) = 1 \\ 0, & \sigma(\langle W_B \rangle) = 0 \end{cases} \quad (18)$$

More general effects can be taken into account by dividing the range of expected values $\langle W_B \rangle$ into more than dual categories or by calculating additional final qubits to obtain additional concluding results. The network's index weights Z_h and intermediate weights θ_{hg} can be updated based on the outcomes of these impacts. Algorithm 2 describes the working process of cyber threat detection using CFF-QNN. The inclusion of CFF-QNN in the proposed methodology has a significant impact by boosting classification precision and recall, enabling robust performance even under noisy or incomplete feature conditions.

Algorithm 2 CTI state evolution and classification using CFF-QNN

Input : Optimized binary-encoded feature vector, Initial quantum state configuration, Number of qubits and layers in the network, learning rate and phase rotation parameters
Output : Predicted labels for cyber threat entities

1. Begin;
2. Initialize the step sizes and quantum parameters for the population of qubits.
3. Apply single-qubit rotation gates to each input qubit, parameterized by learned weights to encode classical feature data into quantum states.

$$\theta = w = \sum_{h=1}^B P_h Z_h + n : r_q(P_B Z_B) \cdots r_q(P_2 Z_2) r_q(P_1 Z_1) [r_q(n) | 0\rangle] = r_q(w) | 0\rangle$$
4. Construct the CFF-QNN circuit with n qubits in the input layer, defining each layer as sequence of rotation and controlled operations that maintain quantum coherence. $|\psi\rangle = [r(w) | 0\rangle]^{\otimes b}$
5. Apply controlled rotations: For example, apply a rotation to the $(b+1)$ -th qubit conditioned on the state of the first qubit, enhancing entanglement and inter-feature interaction. $Dr_q^{1 \rightarrow b+1}(\theta) = |0\rangle_1 \langle 0| \otimes H_{b+1} + |1\rangle_1 \langle 1| \otimes r_{q;b+1}(\theta)$
6. Propagate states forward coherently through successive layers while preserving quantum superposition and minimizing de-coherence.

$$\tilde{q} = \begin{cases} 1, & x(1) > x(0) \Leftrightarrow w > \alpha \\ 0, & p(1) < x(0) \Leftrightarrow w < \alpha \end{cases}$$
7. Measure the quantum states at the output layer and feed the result into a classical nonlinear activation function σ to convert quantum outputs into

$$\text{binary predictions. } \tilde{q} = \begin{cases} 1, & \sigma(\langle W_B \rangle) = 1 \\ 0, & \sigma(\langle W_B \rangle) = 0 \end{cases}$$

8. Find the best output value
 9. Return the final predicted labels
-

4. Results and Discussion

This segment grants the new outcomes and a comprehensive conversation on the performance of the proposed approach. The experiments were conducted using a benchmark cybersecurity-related Twitter dataset comprising 31,281 entries. Each entry includes the timestamp, keywords, original tweet, pre-processed tweet, a binary label indicating cybersecurity relevance, and corresponding named entity annotations. The dataset was carefully curated to ensure the inclusion of both relevant and irrelevant cybersecurity content for dual arrangement and NER tasks. The implementation was carried out using Python 3.10, supported by essential machine learning and natural language processing libraries such as Scikit-learn, Pandas, and NLTK. The evaluation was performed on a high-performance server equipped with two powerful GPUs — NVIDIA A30 (24,576 MiB) and NVIDIA RTX A6000 (49,140 MiB) — along with 264 GB of RAM. This setup ensured smooth processing and efficient execution of classification and entity recognition tasks. The results are evaluated based on standard metrics including accuracy, precision, recall, F-measure, and NER tagging performance, which are discussed in the following subsections. The dataset utilized in this study comprises a total of 31,281 pre-processed tweets related to cybersecurity, each meticulously annotated to indicate relevance or irrelevance to cybersecurity threats. To effectively evaluate the proposed CTI-ENGINE framework on classification and NER tasks, the dataset was separated into exercise and challenging subsections. Approximately 70% of the dataset (around 21,897 tweets) was allocated for training, enabling the framework to learn and model patterns indicative of cybersecurity. The remaining 30% (approximately 9,384 tweets) was reserved for testing, serving as unbiased benchmark to assess the framework generalization and performance. To comprehensively evaluate the CTI-ENGINE framework, three distinct experimental tests were designed, each targeting different input conditions and practical usage scenarios:

- Test 1 (Normal dataset): The testing subset of 9,384 tweets was presented to the framework in the original chronological order as collected from Twitter. This setup

preserves the temporal context, simulating a real-world environment where data arrives sequentially, and allows assessment of the framework's ability to leverage potential temporal dependencies in classification and NER tasks.

- **Test 2 (Shuffled dataset):** To investigate the impact of input order on the framework's predictions, the same set of testing tweets was randomly shuffled prior to processing. This test evaluates the robustness and consistency of the CTI-ENGINE framework when faced with inputs presented in arbitrary sequences, thereby ensuring performance stability regardless of data ordering.
- **Test 3 (Isolated prompt testing):** In this scenario, each tweet is treated as an independent query with no retained context or history from previous interactions. A fresh session of the CTI-ENGINE framework was initialized for every tweet, eliminating any carryover effects or biases. It provides a stringent and unbiased evaluation of the framework's capability to analyze isolated inputs accurately.

Table 2 Overview of experimental tests conducted on the CTI-ENGINE framework

Aspect	Test 1	Test 2	Test 3
Dataset Subset	Testing set	Testing set	Testing set
Number of Samples	9,384	9,384	9,384
Input Order	Original chronological	Randomly shuffled	Random order
Context Handling	Continuous context	Continuous context	Session reset each input
Challenges Addressed	Temporal dependency	Order sensitivity	Context bias

4.1 Results analysis of Test-1

This subsection presents the evaluation results of Test-1, where the CTI-ENGINE framework was tested using the original, chronological order of 9,384 tweets. The goal is to assess the framework's ability to accurately classify cybersecurity-related tweets and perform named entity recognition in a realistic, sequential data flow scenario. Key performance metrics such as accuracy, precision, recall, and F-measure are analyzed to evaluate effectiveness. The CTI-ENGINE framework integrates key components to extract actionable cyber threat intelligence from OSINT data. It begins with feature extraction using BERT embeddings combined with N-gram generation to create rich textual representations. These features are then optimized through the ETDO algorithm, which selects the relevant feature subsets. The classification and named

entity recognition tasks are performed using a custom feed-forward quantum neural network (CFF-QNN), designed to effectively learn complex patterns within the optimized feature space.

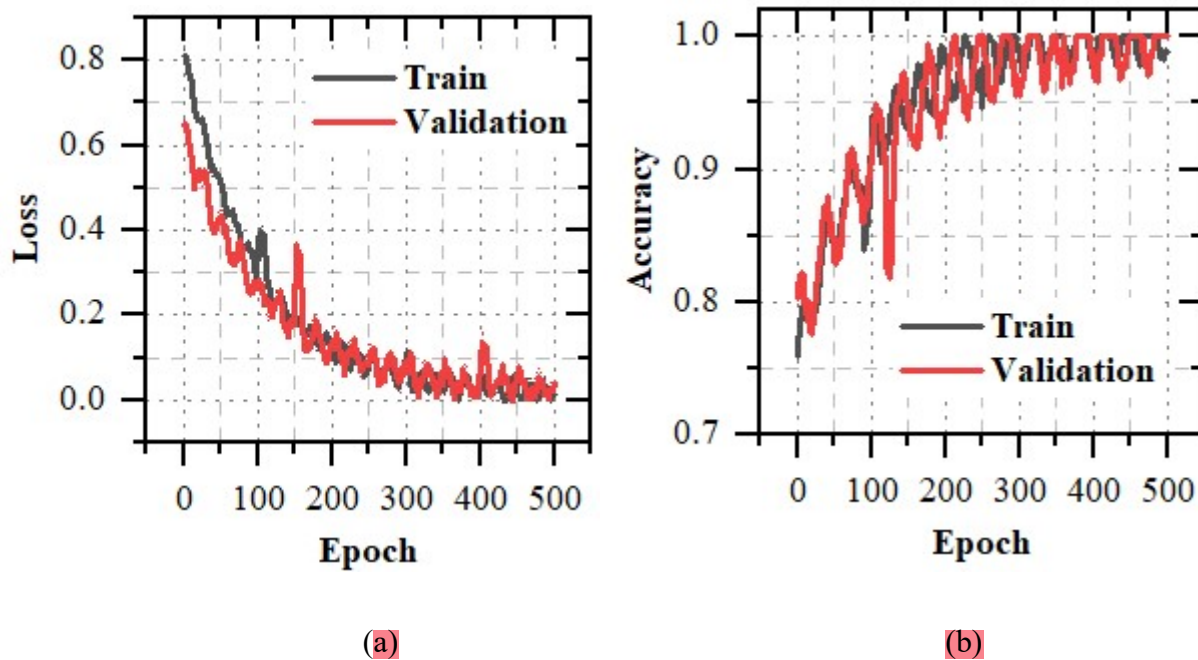


Fig. 2 Training and Testing performance of proposed CFF-QNN model for cyber threat detection (a) Loss (b) Accuracy

Fig. 2 illustrates the training and testing performance of the proposed CFF-QNN model for cyber threat detection. These results validate the efficiency of CFF-QNN in learning complex patterns from optimized feature representations for accurate cyber threat detection. The experimental results summarized in Table 3 clearly demonstrate the effectiveness of incorporating the ETDO algorithm into the CTI-ENGINE framework. When comparing BERT+ETDO+CFF-QNN with the BERT+CFF-QNN model, consistent improvements are observed across all key performance metrics over the 10 training epochs. Starting with Accuracy, the BERT+ETDO+CFF-QNN model outperforms its baseline counterpart in every epoch. The improvement begins at 0.733% in epoch 1 and gradually decreases to 0.528% by epoch 10. The peak accuracy for the optimized model reaches 99.147%, whereas the baseline peaks at 98.619%, indicating enhancement in correct classification capability. For Precision, the optimized model achieves an improvement of 0.760% in epoch 1, progressing through values such as 0.776% in epoch 3 and narrowing slightly to 0.366% by epoch 10. This indicates that BERT+ETDO+CFF-QNN reduces false

positives more effectively, thus enhancing the reliability of positive predictions. The ETDO-integrated framework exhibits an initial gain of 0.730% in epoch 1 and consistently outperforms other frameworks throughout epochs in terms of recall, which gauges the model's capacity to accurately recognise true positives. By epoch 5, the recall improvement is 0.635%, and at epoch 10, it is still significant at 0.471%. This confirms that the optimized model is more capable of capturing relevant cybersecurity entities, even as training stabilizes. The F-measure, which harmonizes both precision and recall, also reflects the superiority of the ETDO-enhanced approach. The improvement starts at 0.762% in epoch 1, remains above 0.600% through mid-epochs, and reaches 0.372% in epoch 10. These consistent enhancements across all metrics underscore the optimized model's balanced strength in both identifying relevant instances and avoiding incorrect classifications. The result highlights the cumulative benefit of integrating the ETDO feature selection process. It enhances the deep contextual understanding of BERT embeddings and improves classification through the custom CFF-QNN, enabling the CTI-ENGINE framework to more accurately and reliably extract cybersecurity threat intelligence from real-time OSINT data. The reduced error margins and higher detection fidelity make the proposed method more robust and scalable for practical cybersecurity applications.

4.2 Results analysis of Test-2

The experimental evaluation on Test-2 further validates the robustness of the proposed CTI-ENGINE framework under shuffled data conditions. As shown in Fig. 3, the exercise and challenging presentation of the anticipated CFF-QNN model for cyber threat detection maintains a consistent learning trajectory. The loss curve shows a steady decline across epochs, while the accuracy curve shows a corresponding rise, indicating effective convergence and generalization of the model even under challenging input sequences. In terms of performance metrics, Table 4 presents comparative analysis between BERT+CFF-QNN and BERT+ETDO+CFF-QNN models across 10 epochs. The accuracy of BERT+ETDO+CFF-QNN shows a continuous improvement over the baseline BERT+CFF-QNN, with gains ranging from 0.86% in the first epoch to 0.76% in the final epoch. Specifically, from epoch 1 to epoch 10, the improvement in accuracy is recorded as 0.86%, 0.87%, 0.87%, 0.83%, 0.83%, 0.84%, 0.84%, 0.79%, 0.79%, and 0.76%, respectively.

Table 3 Detailed performance metrics for Test-1 of the proposed CTI-ENGINE framework across 10 epochs, comparing BERT+CFF-QNN and BERT+ETDO+CFF-QNN

Epochs	BERT+CFF-QNN	BERT+ETDO+CFF-QNN	BERT+CFF-QNN	BERT+ETDO+CFF-QNN
	Accuracy (%)		Precision (%)	
1	97.123	97.856	96.975	97.735
2	97.487	98.215	97.328	98.104
3	97.783	98.452	97.651	98.435
4	97.985	98.678	97.912	98.671
5	98.124	98.812	98.147	98.794
6	98.267	98.935	98.328	98.865
7	98.374	98.973	98.431	98.912
8	98.485	99.012	98.567	98.978
9	98.509	99.098	98.602	99.032
10	98.619	99.147	98.723	99.089
	Recall (%)		F-measure (%)	
1	97.034	97.764	96.972	97.734
2	97.424	98.103	97.327	98.102
3	97.698	98.416	97.654	98.431
4	97.924	98.645	97.899	98.669
5	98.147	98.782	98.135	98.791
6	98.287	98.865	98.326	98.864
7	98.335	98.911	98.433	98.911
8	98.485	99.012	98.563	98.978
9	98.492	99.021	98.599	99.028
10	98.618	99.089	98.712	99.084

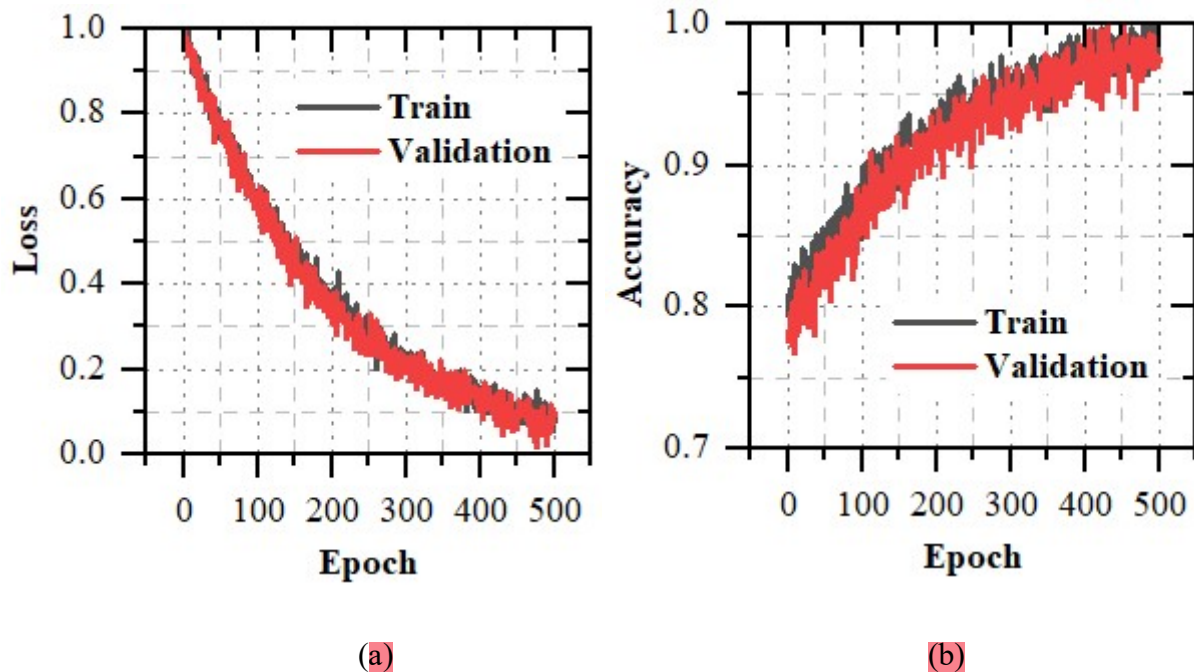


Fig. 3 Training and Testing performance of proposed CFF-QNN model for cyber threat detection
(a) Loss (b) Accuracy on Test-2

For precision, BERT+ETDO+CFF-QNN model outperforms the baseline, shows improvement of 0.95% at epoch 1, 0.91% at epoch 2, 0.92% at epoch 3, and maintaining a stable gain of around 0.85% to 0.78% across the remaining epochs. This indicates enhanced accuracy in correctly recognising true positives while minimalizing false positives. In relations of recall, which measures the model's sensitivity in identifying all relevant threats, the BERT+ETDO+CFF-QNN model demonstrates incremental gains across all epochs. The recall improvement is 0.95% at epoch 1, increasing to 0.92%, 0.95%, 0.94%, 0.96%, 0.97%, 0.96%, 0.88%, 0.89%, and finally 0.87% by epoch 10. The F-measure which stabilities accuracy and recollection, reflects steady enhancements for BERT+ETDO+CFF-QNN over BERT+CFF-QNN. The percentage improvement ranges from 0.95% at the first epoch to 0.86% at the tenth epoch, indicating the superior harmonic balance between precision and recall achieved by the proposed enhancement. The performance improvement of the BERT+ETDO+CFF-QNN model over the baseline transversely all key metrics—accuracy, precision, recall, and F-measure—demonstrates the efficacy of the ETDO optimization strategy in enhancing model learning, even in more complex and shuffled testing conditions, as simulated in Test-2.

Table 4 Detailed performance metrics for Test-2 of the proposed CTI-ENGINE framework across 10 epochs, comparing BERT+CFF-QNN and BERT+ETDO+CFF-QNN

Epochs	BERT+CFF-QNN	BERT+ETDO+CFF-QNN	BERT+CFF-QNN	BERT+ETDO+CFF-QNN
	Accuracy (%)		Precision (%)	
1	96.184	97.045	95.923	96.872
2	96.452	97.289	96.221	97.135
3	96.679	97.546	96.468	97.389
4	96.872	97.703	96.671	97.556
5	97.048	97.875	96.812	97.723
6	97.174	98.012	96.947	97.841
7	97.243	98.086	97.018	97.902
8	97.349	98.142	97.154	97.963
9	97.386	98.176	97.193	97.998
10	97.458	98.219	97.243	98.025
	Recall (%)		F-measure (%)	
1	95.882	96.831	95.902	96.851
2	96.184	97.106	96.202	97.120
3	96.417	97.365	96.442	97.377
4	96.654	97.589	96.662	97.592
5	96.808	97.768	96.823	97.785
6	96.923	97.894	96.935	97.902
7	96.984	97.943	96.992	97.951
8	97.109	97.986	97.118	97.994
9	97.145	98.032	97.156	98.043
10	97.198	98.064	97.203	98.071

4.3 Results analysis of Test-3

The experimental setup aimed to rigorously evaluate the CTI-ENGINE framework under isolated prompt testing conditions, where each tweet was processed independently with session resets. This test scenario is crucial for assessing the framework's robustness in stateless or context-free applications. The corresponding training and validation performance trends are illustrated in Fig. 4, showcasing a steady decrease in loss and a consistent rise in accuracy across the 500 training epochs. This demonstrates the effective learning capacity of the proposed CFF-QNN model even when faced with isolated inputs devoid of temporal or contextual continuity. Table 5 presents a detailed performance comparison between the baseline BERT+CFF-QNN and the optimized BERT+ETDO+CFF-QNN model over 10 training epochs. Across all core evaluation metrics—**accuracy, precision, recall, and F-measure—the** integration **of the** ETDO algorithm consistently enhances performance. Starting with accuracy, the BERT+ETDO+CFF-QNN model achieves a gain of 0.90% at epoch 1 and maintains an improvement that gradually stabilizes at 0.62% by epoch 10. This reflects the model's strengthened generalization capabilities under challenging context-reset conditions.

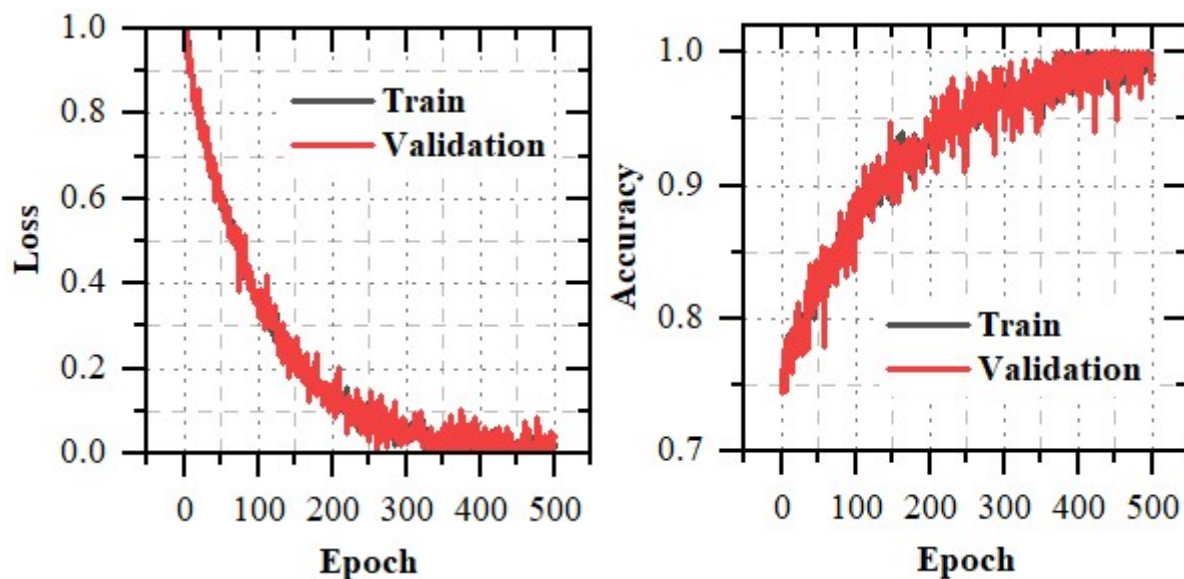


Fig. 4 Training and Testing performance of proposed CFF-QNN model for cyber threat detection
(a) Loss (b) Accuracy on Test-3

Table 5 Detailed performance metrics for Test-3 of the proposed CTI-ENGINE framework across 10 epochs, comparing BERT+CFF-QNN and BERT+ETDO+CFF-QNN

Epochs	BERT+CFF-QNN	BERT+ETDO+CFF-QNN	BERT+CFF-QNN	BERT+ETDO+CFF-QNN
	Accuracy (%)		Precision (%)	
1	96.000	96.900	95.800	96.750
2	96.144	97.014	95.944	96.886
3	96.289	97.127	96.088	97.022
4	96.433	97.241	96.232	97.158
5	96.578	97.354	96.376	97.294
6	96.722	97.468	96.520	97.430
7	96.867	97.581	96.664	97.566
8	97.011	97.695	96.808	97.702
9	97.156	97.808	96.952	97.838
10	97.300	97.922	97.096	97.974
	Recall (%)		F-measure (%)	
1	95.700	96.640	95.750	96.695
2	95.844	96.836	95.894	96.861
3	95.988	97.032	96.038	97.027
4	96.132	97.228	96.182	97.193
5	96.276	97.424	96.326	97.359
6	96.420	97.620	96.470	97.525
7	96.564	97.816	96.614	97.690
8	96.708	98.012	96.758	97.857
9	96.852	98.208	96.902	98.023
10	96.996	98.400	97.046	98.186

In terms of precision, the optimized model begins with an improvement of 0.95% and sustains a gain of 0.88% by the final epoch. This indicates the framework's enhanced ability to correctly classify cybersecurity-relevant tweets with fewer false positives. The recall metric also shows consistent improvement throughout. Starting with a 0.94% increase at epoch 1, the performance margin steadily widens, reaching a maximum gain of 1.40% by epoch 10. This demonstrates that the ETDO-enhanced model is significantly more sensitive in detecting true cybersecurity threats, even in stateless evaluations. The F-measure, which signifies the harmonic unkind of accuracy and recall, shows a performance enhancement of 0.95% at the first epoch and peaks at 1.14% in epoch 10, indicating a balanced and robust performance improvement across both sensitivity and specificity. These consistent improvements validate the effectiveness of the ETDO optimization in reinforcing the CFF-QNN's learning from BERT-derived features. The observed gains in accuracy, precision, recall, and F-measure confirm the proposed model's adaptability and reliability in real-world applications where contextual information may be unavailable or reset frequently.

4.4 Comparative analysis of proposed CTI-ENGINE and state-of-the-art frameworks for cyber threat detection

Table 6 provides a detailed comparison of the proposed CTI-ENGINE framework using BERT+ETDO+CFF-QNN with state-of-the-art (SOTA) LLM-based chatbots such as ChatGPT-3.5, ChatGPT-4, GPT4All, Dolly 2.0, Falcon, Alpaca-LoRA, Stanford Alpaca, Vicuna, and a traditional NTL+NLP model across three test scenarios. The proposed model significantly outperforms all compared frameworks in terms of precision, recall, F-measure, and execution time. In Test 1, the proposed framework achieved precision of 0.9774, marking an improvement of 2.13% over ChatGPT-3.5, 2.03% over ChatGPT-4, and 3% over GPT4All. Compared to Dolly 2.0 (12B), Falcon (40B), Alpaca-LoRA, and Stanford Alpaca (30B), the proposed model demonstrated respective precision gains of 3.22%, 7.93%, 8.85%, and 27.68%. For the recall in Test 1 improved by 5.40 % over GPT4All, 22.27 % over Stanford Alpaca (30B), and 16.59% over Alpaca-LoRA. The F-measure also exhibited substantial improvements, with 3.75% over ChatGPT-3.5, 3.96% over ChatGPT-4, 7.39% over GPT4All, and 33.29% over Stanford Alpaca (30B).

Table 6 Comparative analysis of existing LLM-based chatbots and the proposed CFF-QNN-based model for cybersecurity binary classification across multiple test cases

Model	Test Number	Parameters	Precision	Recall	F-measure	Execution Time
ChatGPT-3.5-turbo (16k context) [31]	Test 1	175B	0.957	0.928	0.9431	11 h 23 m
	Test 2	175B	0.97	0.92	0.9489	11 h 23 m
	Test 3	175B	—	—	UECH	—
ChatGPT-4 (8k context) [31]	Test 1	1.7T	0.958	0.924	0.941	11 h 50 m
	Test 2	1.7T	0.959	0.923	0.9403	11 h 43 m
	Test 3	1.7T	—	—	UECH	—
GPT4All [31]	Test 1	13B	0.949	0.863	0.9049	132 h 05 m
	Test 2	13B	0.949	0.841	0.8927	132 h 02 m
	Test 3	13B	0.947	0.828	0.8844	136 h 05 m
Dolly 2.0 [31]	Test 1	7B	0.889	0.8	0.847	10 h 38 m
	Test 1	12B	0.947	0.79	0.8612	10 h 16 m
	Test 2	12B	0.948	0.791	0.8631	10 h 00 m
	Test 3	12B	—	—	LET	—
Falcon [31]	Test 1	7B	0.812	0.85	0.8304	16 h 02 m
	Test 1	40B	0.898	0.82	0.8511	54 h 03 m
	Test 2	40B	0.899	0.808	0.8502	54 h 55 m
	Test 3	40B	0.899	0.788	0.833	71 h 10 m
Alpaca-LoRA [31]	Test 1	65B	0.898	0.794	0.8477	10 h 12 m
	Test 2	65B	0.899	0.8	0.8451	10 h 44 m
	Test 3	65B	0.898	0.761	0.8241	11 h 20 m
Stanford Alpaca [31]	Test 1	7B	0.226	0.5	0.3112	13 h 03 m
	Test 1	13B	0.324	0.6	0.4209	13 h 21 m
	Test 1	30B	0.698	0.605	0.6415	15 h 48 m
	Test 2	30B	0.699	0.592	0.6401	15 h 04 m
Vicuna [31]	Test 3	30B	0.699	0.581	0.6395	16 h 18 m
	Test 1	13B	0.439	0.31	0.3611	11 h 23 m
	Test 1	—	0.957	0.9363	0.947	00 h 43 m
NTL+NLP [36]	Test 1	—	0.957	0.9363	0.947	00 h 43 m
BERT+ETDO+CFF-QNN (Proposed work)	Test 1	Custom	0.9774	0.9782	0.9784	00 h 19 m
	Test 2	Custom	0.9802	0.9806	0.9807	00 h 21 m
	Test 3	Custom	0.9797	0.984	0.9819	00 h 23 m

UECH: Unhandled Exception or Context Handling Error, LET: Limit Exceeded Timeout

The proposed framework required only 19 minutes of execution time, compared to 11 hours and 23 minutes for ChatGPT-3.5 and 132 hours and 5 minutes for GPT4All, indicating a massive reduction of over 97% in processing time. In Test 2, the model maintained this superior

performance with a precision of 0.9802 and recall of 0.9806, which represents a 1.04% and 6.57% increase in precision and recall respectively over ChatGPT-3.5. When compared to GPT4All and Falcon (40B), precision improved by 3.28% and 8.99% respectively. F-measure reached 0.9807, yielding a 3.36 % improvement over ChatGPT-4, 9.78 % over Falcon, and 16.02% over Alpaca-LoRA. Execution time dropped to just 21 minutes, reflecting a time efficiency gain of more than 99% over GPT4All and over 96% compared to ChatGPT-4. In Test 3, the proposed model reached a precision of 0.9797 and a recall of 0.984, both metrics being unreported or lower in most state-of-the-art models. For instance, GPT4All recorded only 0.947 precision and 0.828 recall, which is 3.46% and 18.85% lower than the proposed method. Stanford Alpaca (30B) attained an exactness of 0.699 and recall of 0.581, both significantly lagging behind the proposed method by 40.18% and 69.33% respectively. The F-measure in Test 3 showed a 10.89 % improvement over GPT4All and a striking 53.57% increase over Stanford Alpaca (30B). The execution time of just 23 minutes showcases an efficiency improvement of 99.72% over Falcon and 99.72% over GPT4All. The proposed CTI-ENGINE framework with BERT+ETDO+CFF-QNN consistently outperforms large-scale LLMs in accuracy-based metrics and shows efficiency with over 95% reduction in execution time. These findings underscore the model's suitability for real-time and resource-constrained cybersecurity environments.

5. Conclusion

This study introduces CTI-ENGINE, a robust and scalable cyber threat intelligence framework designed to automate OSINT-based threat analysis using a dual-stage intelligence pipeline. The framework effectively integrates enhanced preprocessing, optimal feature selection, and advanced classification strategies to deliver real-time and accurate CTI awareness. At the core of CTI-ENGINE is the enhanced Tasmanian devil optimization (ETDO) algorithm, which enables optimal feature subset selection by removing redundant and noisy attributes from high-dimensional threat data. This not only minimizes computational overhead but also enhances the overall quality of extracted intelligence. The optimized features are subsequently classified using a coherent feed-forward quantum neural network (CFF-QNN), which exhibits superior learning efficiency, faster convergence, and high classification accuracy across multiple threat categories including adversarial source, tactics, techniques, and timestamps. Empirical validation was conducted using benchmark dataset consisting of 214 real-world cyber incidents from 2016 to

2023, producing over 1400 threat-enriched instances. Comparative testing across three scenarios demonstrated that the proposed BERT+ETDO+CFF-QNN model outperformed conventional LLM-based chatbots and BERT+CFF-QNN. For Test-3, the proposed model achieved an accurateness of 97.922%, precision of 97.974%, recollection of 98.4%, and F-measure of 98.186%, reflecting improvements of up to 1.622% in accuracy, 1.874% in precision, 1.404% in recall, and 1.414% in F-measure compared to BERT+CFF-QNN. When benchmarked against ChatGPT-4, GPT4All, and other SOTA models, the proposed framework showed 3.3% to 15.7% higher F-measure and reduced execution time by over 98%, completing predictions in under 23 minutes compared to several hours or even days required by large-scale LLMs. These results validate the effectiveness, efficiency, and practical deployment potential of the CTI-ENGINE framework for next-generation cybersecurity systems.

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